

JUDGE DOCUMENTATION

Confidential Payroll

Privacy-preserving payroll, treasury, compliance, and private board governance built with Zama fhEVM.

CATEGORY

Confidential on-chain enterprise operations

CORE TECHNOLOGY

Zama fhEVM + relay SDK

PRIMARY OUTCOME

Execute payroll logic without exposing salaries publicly

Abstract

Confidential Payroll is a privacy-first payroll and governance platform built on Ethereum-compatible infrastructure with Zama's fully homomorphic encryption stack. The project addresses a core contradiction in enterprise blockchain adoption: organizations want the auditability, automation, and composability of smart contracts, but they cannot expose salaries, treasury balances, board votes, or compliance decisions on a public ledger. Our solution keeps sensitive business data encrypted on-chain while still allowing contracts to execute payroll logic, enforce compliance rules, manage treasury state, and run private board voting.

Zama is the core enabling technology in this system. We use Zama fhEVM contracts to store and compute over encrypted values such as employee salaries, treasury balances, aggregated payroll totals, and governance tallies. On the frontend, we use Zama's relay SDK to encrypt inputs in the browser and request authorized decryptions only for eligible users.

Problem Statement

Traditional payroll systems are private but centralized. Conventional on-chain systems are programmable but transparent by default. This leaves organizations with a difficult tradeoff: either accept opaque Web2 infrastructure, or expose sensitive compensation and governance data to the public.

We focused on three concrete problems:

1. Salary confidentiality is lost in transparent smart contract systems.
2. Treasury and internal financial activity become visible on public blockchains.
3. Board decisions and internal approvals cannot be executed privately in standard on-chain workflows.

Core thesis: enterprise payroll should gain blockchain automation without turning compensation, treasury state, and board behavior into public data.

Solution Overview

Confidential Payroll combines smart contract automation with encrypted inputs and encrypted contract state. The project supports payroll management, treasury logic, compliance gating, and private governance in a single privacy-preserving workflow.

- Employers add employees with encrypted salaries.
- Payroll is executed through on-chain logic rather than off-chain bookkeeping.
- Treasury values remain encrypted instead of being readable on-chain balances.
- Compliance policies can gate payroll actions.
- Board members vote privately, and only the final aggregate outcome is revealed.

How We Used Zama FHE

Contract Layer

In the payroll contract, sensitive values such as salaries, employee balances, treasury balance, and payroll aggregates are stored and processed as encrypted integers. In the governance contract, encrypted ballots are converted into encrypted tallies so that no one can inspect individual votes during the voting period.

Frontend Layer

The frontend uses Zama's relay SDK to initialize the fhEVM environment, encrypt values in the browser, submit encrypted handles and proofs to contracts, and let authorized users request decryption only for values they are allowed to access.

Architecture

- Frontend: React, TypeScript, ethers.js, Zama relay SDK
- Smart contracts: ConfidentialPayroll.sol and ConfidentialGovernance.sol
- Execution environment: Ethereum Sepolia with Zama-compatible encrypted execution flow

Workflows

Employer

The employer deploys the payroll contract, sets compliance policy if needed, adds employees with encrypted salaries, and runs payroll cycles without publishing salary values in plaintext.

Employee

The employee connects to the shared payroll contract, decrypts their own authorized salary and claimable balance, and withdraws funds through the contract.

Governance

The admin deploys a governance contract, adds board members, creates proposals, and collects encrypted votes. During the vote, no individual preference is revealed. After finalization, the aggregate result becomes available.

Key Problems Solved

- Protecting employee salary privacy on a public blockchain
- Preserving treasury confidentiality in payroll operations
- Supporting private internal approvals and board decisions
- Combining compliance controls with encrypted financial workflows

Why This Matters

This project demonstrates that on-chain enterprise systems do not have to choose between automation and confidentiality. Zama fhEVM makes it possible to build business applications where sensitive data remains encrypted throughout the smart contract lifecycle while still benefiting from programmable execution.

Roadmap

1. Bind governance outcomes directly to payroll actions such as hiring approvals, salary updates, and role appointments.
2. Expand auditor and tax authority workflows with privacy-preserving reporting.
3. Strengthen enterprise policy configuration and role management.
4. Add richer organization-level analytics based on encrypted aggregates.

Confidential Payroll documentation prepared for hackathon and judge review. Repository link and live demo link can be attached alongside this document.